

STICK FOR The Blind

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A stick for the blind proposed here is an assistive device that helps visually impaired individuals navigate safely. It features an infrared (IR) sensor that detects obstacles and activates a buzzer to alert the user. A 9V battery powers the circuit for convenient operation.

Lightweight and portable, the stick enhances mobility by providing real-time feedback on surroundings. It integrates an IR sensor module, a PNP transistor, and a piezo buzzer to detect obstacles and issue audible alerts. Compact yet highly effective, the circuit improves safety for visually impaired individuals.

The IR sensor module operates by emitting and receiving infrared signals. When it detects an obstacle, the transistor amplifies the signal, triggering the buzzer. This precise and efficient system enhances user confidence and independence. Fig. 1 shows the author's prototype.

Circuit and working

Fig. 2 shows the circuit diagram of the stick for the blind. It is built

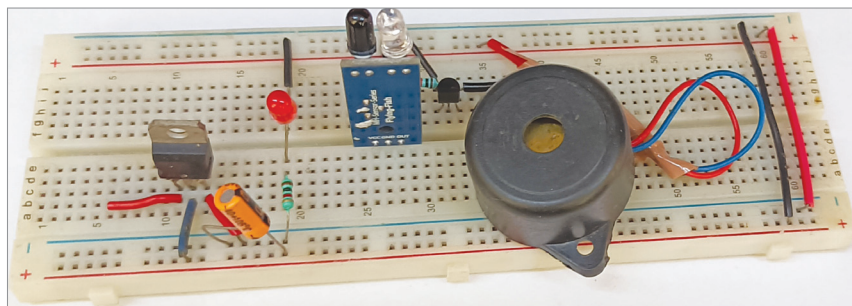


Fig. 1: Author's prototype

around an IR sensor module connected across CON2, a PNP transistor BC558 (T1), a buzzer (PZ1), and a few other components. A 9V battery is sufficient to power the circuit.

The 9V battery is connected across CON1. A voltage regulator, LM7805, reduces the 9V to 5V to power the circuit, as the IR module operates at around 5V. Switch S1 is used to turn on the circuit, and when LED1 lights up, it indicates that the circuit is ready for use.

The heart of the circuit is the IR module, which detects objects within a specific distance. The module is fitted at the bottom of the stick. When an obstacle is detected, the stick signals the user by producing a sound through the buzzer.

An IR module can transmit and receive infrared data. In other words, it houses an IR transmitter and a receiver in a single unit with shared circuitry. The IR sensor module has three pins—VCC, GND, and OUT—as shown in Fig. 3. The white LED in the IR module acts

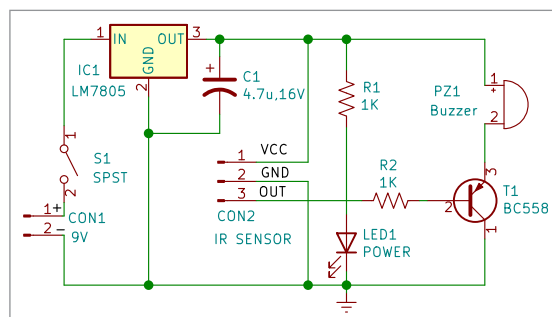


Fig. 2: Circuit diagram

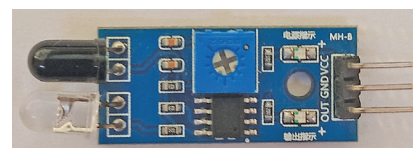


Fig. 3: IR sensor module

as a transmitter, while the black LED (photodiode) is a receiver. The IR transmitter continuously emits infrared light, and the IR receiver checks for reflected light.

If the light reflects off an object in front, the IR receiver detects it, allowing the IR sensor module to identify the object. The module has an inbuilt potentiometer to adjust the range. It also features two inbuilt SMD LEDs—one for power, which remains on during operation, and another for signal detection, which turns on when an object is

PARTS LIST

Semiconductors:

- IC1 - LM7805, 5V voltage regulator
- T1 - BC558 PNP transistor
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2 - 1k Ω

Capacitors:

- C1 - 4.7 μ F, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- CON2 - 3-pin connector
- S1 - SPST switch
- PZ1 - Buzzer
- IR sensor module
- 9V DC supply/9V battery with connector
- Normal stick
- Wires and screws